

Fired Heater engineering manual

Estimate absorbed heat and fuel efficiency, compare the process-side duty, and track combustion losses.

Run the Python model

Open fired_heater_engine.py in Python IDLE. Select Run, then Run Module (F5). Python 3.10 or newer is required; no extra packages or other files are needed.

The built-in example writes a results JSON file beside the Python file. Edit SETTINGS near the top and run again to change assumptions. To use your own measurements, run the file with an input JSON filename. Inputs use the fields and units below.

The input envelope contains schemaVersion, model, source, timezone, config and measurements. Each measurement contains timestamp, values and optional quality. Use ISO timestamps with an offset. Bad, stale and missing signals remain unavailable. The AVEVA mapping is provisional; a live connection has not been verified.

Inputs and units

Model inputs · 7 rows		
INPUT	UNIT	USE
Timestamp; fuel mass flow	date/time; kg/s	Fuel heat input and trends.
Fuel composition and LHV	Mole fractions; MJ/kg	Select a saved fuel or enter a custom blend in Configuration.
Fuel, combustion-air and stack temperature	°C	Sensible heat and stack loss.
Stack oxygen	vol% wet	Combustion-air balance, matching the supplied heater example.
Process flow, heat capacity and inlet/outlet temperature	kg/s; kJ/(kg·K); °C	Optional independent process-side duty.
Bridgewall temperatures	°C	Average available readings and compare with configured limits.
Reference temperature, radiation loss and design duty	°C; %; kW	Heat-balance assumptions.

Calculations

FUEL BLEND

$$MW = \sum x_i MW_i; LHV = \sum w_i LHV_i$$

Custom mole fractions are normalized; mass fractions w_i follow from molecular weights.

COMBUSTION AIR

$$O_{2\text{excess}} = y \times n_{\text{wet,stoich}} / [1 - y \times (1 + N_2/O_{2\text{air}})]$$

y is wet oxygen fraction. Complete combustion sets CO_2 , water and stoichiometric oxygen; air uses 21% O_2 and 79% N_2 by volume.
Flue mass = fuel + air.

HEAT BALANCE

$$Q_{\text{absorbed}} = \dot{m}_{\text{fuel}} \times LHV + Q_{\text{fuel,sensible}} + Q_{\text{air,sensible}} - Q_{\text{stack}} - Q_{\text{radiation}}$$

kW with LHV in kJ/kg. Each sensible term is $\dot{m}C_p(T - T_{\text{reference}})$. Radiation is a configured fraction of fuel LHV input.

EFFICIENCY

$$\eta_{\text{balance}} = 100 \times Q_{\text{absorbed}} / Q_{\text{LHV}}; Q_{\text{process}} = \dot{m}_{\text{process}} C_p \Delta T$$

$$\text{Process efficiency} = 100 \times Q_{\text{process}} / Q_{\text{LHV}}. \text{ Closure} = 100 \times |Q_{\text{absorbed}} - Q_{\text{process}}| / Q_{\text{LHV}}.$$

LOSS ESTIMATE

$$\eta_{\text{loss}} = 100 - \text{dry-gas loss} - \text{moisture loss} - \text{radiation loss} - \text{unaccounted loss}$$

Separate simplified estimate using constant heat capacities. Excess air = $100 \times (\text{actual air} / \text{stoichiometric air} - 1)$, using the same wet-oxygen composition balance. This is not a full PTC 4 test.

Useful trends

- Absorbed and process-side duty versus design, with closure discrepancy.
- Efficiency alongside stack temperature and oxygen to distinguish changes in heat loss and excess air.
- Fuel input, combustion-air flow, dry-gas/moisture losses and bridgewall temperature.

Assumptions and limits

- Complete combustion, constant heat capacities and fixed heat-loss assumptions. CO, SO₂, air leakage and steam-injection enthalpy are not explicitly resolved.
- Do not enter dry-basis oxygen as wet-basis oxygen. The simplified loss estimate uses a different approximation and will not exactly match the composition balance.
- Process duty assumes sensible heating with the entered Cp; phase change requires an enthalpy-based model. Alerts use configured limits, not universal API or ASME thresholds.

Measured values come from instruments or samples. Calculated values use those inputs and the configured assumptions. Design values are references. Simulated values demonstrate the model and are not plant measurements. Missing results remain unavailable.

Sources

- Supplied Furnace Heat and Material Balances workbook and Furnace efficiency example: wet oxygen and heat-balance structure
- ASME PTC 4: fired steam-generator test framework, not certification of this simplified calculation